

Multiple Choice (20 marks)

1. As the number of training examples goes to infinity, your model trained on that data will have:
 - (a) Lower variance
 - (b) Higher variance
 - (c) Same variance
 - (d) None of the above
2. Statement 1— Highway networks were introduced after ResNets and eschew max pooling in favor of convolutions. Statement 2— DenseNets usually cost more memory...
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. What is the dimensionality of the null space of the following matrix? $A = \begin{bmatrix} 3 & 2 & -9 \\ -6 & -4 & 18 \\ 12 & 8 & -36 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
4. Another term for out-of-distribution detection is?
 - (a) anomaly detection
 - (b) one-class detection
 - (c) train-test mismatch robustness
 - (d) background detection
5. Neural networks:
 - (a) Optimize a convex objective function
 - (b) Can only be trained with stochastic gradient descent
 - (c) Can use a mix of different activation functions
 - (d) None of the above
6. The number of test examples needed to get statistically significant results should be _

- (a) Larger if the error rate is larger.
 - (b) Larger if the error rate is smaller.
 - (c) Smaller if the error rate is smaller.
 - (d) It does not matter.
7. If N is the number of instances in the training dataset, nearest neighbors has a classification run time of
- (a) $O(1)$
 - (b) $O(N)$
 - (c) $O(\log N)$
 - (d) $O(N^2)$
8. Statement 1— A neural network's convergence depends on the learning rate. Statement 2— Dropout multiplies randomly chosen activation values by zero.
- (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
9. For polynomial regression, which one of these structural assumptions is the one that most affects the trade-off between underfitting and overfitting:
- (a) The polynomial degree
 - (b) Whether we learn the weights by matrix inversion or gradient descent
 - (c) The assumed variance of the Gaussian noise
 - (d) The use of a constant-term unit input
10. Existential risks posed by AI are most commonly associated with which of the following professors?
- (a) Nando de Freitas
 - (b) Yann LeCun
 - (c) Stuart Russell
 - (d) Jitendra Malik

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Which one of the following is the main reason for pruning a Decision Tree?' is 'To avoid overfitting the training set'.
2. True or False: The correct answer to 'Statement 1— RoBERTa pretrains on a corpus that is approximate 10x larger than the corpus BERT pretrained on. Statement 2— ResNeXts in...' is 'False, False'.
3. True or False: The correct answer to 'Statement 1— Overfitting is more likely when the set of training data is small. Statement 2— Overfitting is more likely when the...' is 'False, True'.
4. True or False: The correct answer to 'MLE estimates are often undesirable because' is 'they are biased'.
5. True or False: The correct answer to 'Which of the following is NOT supervised learning?' is 'PCA'.
6. True or False: The correct answer to 'Which among the following prevents overfitting when we perform bagging?' is 'The use of classification algorithms which are not prone to overfitting'.
7. True or False: The correct answer to 'Statement 1— The softmax function is commonly used in multiclass logistic regression. Statement 2— The temperature of a nonuniform...' is 'True, True'.
8. True or False: The correct answer to 'Statement 1— Industrial-scale neural networks are normally trained on CPUs, not GPUs. Statement 2— The ResNet-50 model has over 1 billion...' is 'False, False'.
9. True or False: The correct answer to 'Predicting the amount of rainfall in a region based on various cues is a ----- problem.' is 'Supervised learning'.
10. True or False: The correct answer to 'Statement 1— The original ResNets and Transformers are feedforward neural networks. Statement 2— The original Transformers use self-...' is 'True, False'.

End of Paper